

Exercise 4

Add a *Members* column to the `BandAdmin` listing page that contains multiple links, one to each `Musician` in the `Band`. Note that appending to a string makes it unsafe again. Use `mark_safe` to fix that.

To show the members associated with a band, you need to create and display a custom column that contains a link to the `MembersAdmin` page for each associated musician. The new version of `BandAdmin` is shown in the following listing.

Listing E.21 `BandAdmin` class with links to associated musicians

```
# RiffMates/bands/admin.py
...
@admin.register(Band)
class BandAdmin(admin.ModelAdmin):
    list_display = ("id", "name", "show_members")
    search_fields = ("name",)

    def show_members(self, obj):
        members = obj.musicians.all()
        links = []

        url = reverse("admin:bands_musician_changelist")

        for member in members:
            parm = f"?id={member.id}"
            link = format_html(
                '<a href="{}">{</a>', url, parm, member.last_name
            )
            links.append(link)

        return mark_safe(", ".join(links))

    show_members.short_description = "Members"
```

Look up the `MusicianAdmin` page's URL.

A custom column to display links to musicians

Query all `Musician` objects associated with the `Band` being processed.

Each musician results in its own link, which will be stored in this list.

Use `format_html()` to create a link to each musician's Admin page.

Join the link data into a single string. Processing the strings marks them as unsafe again, so they need to be passed to `mark_safe()` to ensure the HTML is displayed unescaped by the Django Admin.

E.5 Chapter 6

Chapter 6 introduced users and user management, including how to store data associated with a user and deal with passwords.

Exercise 1

The `@login_required` decorator only checks if a user got authenticated. A more generic view restriction can be performed using the `@user_passes_test` decorator in the same library. It takes one required argument: a reference to a function that takes the `User` object as an argument and returns `True` if the user should be allowed